

Persuasive System Design: State of the Art and Future Directions

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ABSTRACT

This paper provides an overview of the current state of the art in persuasive systems design. All peer-reviewed full papers published at the first three International Conferences on Persuasive Technology were analyzed employing a literature review framework. Results from this analysis are discussed and directions for future research are suggested. Most research papers so far have been experimental. Five out of six of these papers (84.4%) have addressed behavioral change rather than an attitude change. Tailoring, tunneling, reduction and social comparison have been the most studied methods for persuasion. Quite, surprisingly ethical considerations have remained largely unaddressed in these papers. In general, many of the research papers seem to describe the investigated persuasive systems in a relatively vague manner leaving room for some improvement.

Categories and Subject Descriptors

H.1.2 [Models and Principles]: User/Machine Systems – human factors, software psychology. H.5.2 [Information Interfaces and Presentation (e.g. HCI)]: User Interfaces – theory and methods, user-centered design.

General Terms

Design, Human Factors

Keywords

Literature analysis, persuasive systems, persuasive technology, persuasive design, behavior change, attitude change, sustainability, interaction design, design methods.

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1. INTRODUCTION

This paper investigates the current body of research knowledge about persuasive systems produced during 2006-2008. This is accomplished through a structured literature review and analysis based on the Persuasive Technology conferences. This conference series is the only well-established scientific forum dedicated to persuasive systems design so far. It was deemed appropriate to investigate the field now, since the first three years of conferences provide a critical mass of research for review. The aim of this analysis is to recognize the predominant research themes and methodologies as well as the design approaches in the field.

In using technology as a vehicle of persuasion, we touch upon a central part of being human, namely intentional communication. Whenever we communicate deliberately with a clear purpose and outcome in mind, we are engaging in persuasion. This is not new; but building ‘machines’ that conduct persuasion on our behalf is. When Aristotle in 400 B.C. defined rhetoric as “...the faculty of observing in any given case the available means of persuasion” [1], he could not have dreamed that one day such a faculty would include deploying systems designed to infuse cognitive and/or emotional changes, in essence delegating the persuasion itself to be conducted by machines. From a historic perspective, such conduct has only recently become possible [2]. As society at large continues to adopt systems that persuade, it becomes increasingly relevant for us to understand how to design such technologies and how to analyze them. In dealing with technology-mediated persuasion, we venture into an area that sits at the intersection of many disciplines and that can be studied from many diverse viewpoints; this is due to the omnipotent nature of persuasion.

At least four recognized key computer-based fields of research for persuasive systems and design can be recognized: human-computer interaction, computer-mediated communication, information systems, and affective computing. We should also observe two key disciplines from the humane sciences namely psychology and rhetoric, as they address core aspects of human persuasion and the cognitive traits that apply within that area.

Human-computer interaction as a discipline is concerned with enabling humans to design computer systems that are usable and

understandable; including how information can be gathered for such designs. The object is to create cognitive models that allow for users to interact (or communicate) with machines so that we may operate them [3, 4].

Computer-mediated communication investigates how technology affects our modes of communications, what happens when a message travels via computers between humans, and how the technology transmission impacts the communication. This includes the study of synchronous and asynchronous communication.

Information systems approach treats the software, databases, and the content provided for the user as a systemic whole and it emphasizes usefulness, i.e. the organizational and end-user benefit to be reaped from this. Many of these systems are developed for work contexts. [5].

Affective computing helps recognize, interpret, and process human emotions. Thus, the aim is to create systems that can express affect but also perceive affect; in essence creating systems that can interpret the emotional state of its human users and also change the machine response accordingly thus presenting a fitting response [6].

Psychology as an academic discipline is concerned with the study of human behavior by examining mental processes. It describes important cognitive traits of humans, their perception and their emotions. Among other things, the study of human motivation resides in the field of psychology. Traditionally cognitive psychology has played a large part in computer interface design as such [7].

Rhetoric holds a special place in the study of computer-mediated persuasion [8, 9], as rhetoric is the father of both modern humanistic thinking and philosophy coining the term persuasion itself [1, 10, 11]

The disciplines mentioned above overlap and are very broad areas. However, we must also note that none of these *on their own* encapsulate technology-mediated persuasion; rather they seem to touch upon some aspect of it. What distinguishes persuasive systems from other systems is that persuasive systems are inherently transformative. Persuasive systems deliberately attempt to infuse a cognitive and/or an emotional change in the mental state of a user to transform the user's current cognitive state into another planned state. The focus of any persuasive system must be a *technology-mediated transformation of either attitudes or behaviors, including a transformation by bolstering or reinforcing existing attitudes or behaviors* [12, 13].

The paper falls in three sections: Firstly, we will present the selected research method and describe the framework for literature review as well as the selection of the papers included in the review. Secondly, we will explain our gathering of data, the analysis, and the findings. Thirdly, we will outline the implications and future directions for the researchers and practitioners in the field.

2. Research method

It would have been possible to search for research papers published in a broad range of journals, e.g. in the fields described in previous section, that that would touch upon individual elements of persuasive technology. However, the annual Persuasive Technology conferences represent the most coherent

body of research knowledge on the topic. To ensure the quality of the research work, we included only full peer-reviewed papers for the analysis. Notes, short papers and posters may naturally hold both interesting and valid knowledge, but oftentimes they describe work-in-progress. We also omitted invited keynote papers since they had not undergone the same type of review process as the submitted full papers. This brought the total body of papers to be reviewed to 51 [17, 18, 19] (2006: 13, 2007: 21, and 2008: 17).

The literature review was conducted by employing the Persuasive Systems Design model (hereafter the PSD Model) [12] as the framework. This model provides a recent and extensive conceptualization of technology-mediated persuasion. While the model was originally created to prescribe persuasive designs and software requirements, it also supports categorizing and mapping of persuasive elements which makes it applicable for the literature analysis. The model was originally presented at the Persuasive Technology 2008 conference [13], and an extended version of it has been published in the Communications of the Association for Information Systems [12]. The latter version was used for the work presented here. To compensate for any shortcomings in the model, we used some additional variables. A total of 35 variables were extrapolated from the PSD model and adopted for the literature review framework and four variables were added for further clarity.

2.1 The PSD Model: Persuasion Context

In the PSD model, the *Persuasion context* comprises elements influencing whether persuasion can take place. The context itself is composed of the intent, event, and strategy. See Fig. 1.

The Persuasion Context:

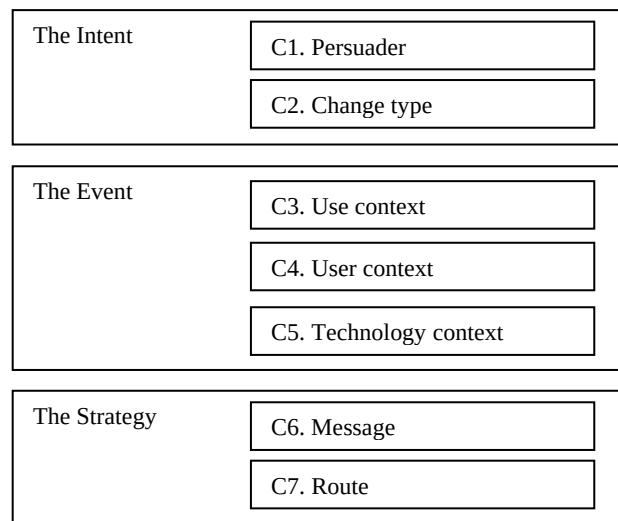


Figure 1. Core components of the PSD Model [12].

The intent: The intent includes the *persuader* (C1, C to denote Context) and the deliberate target behavior that the system is to cause in the user (C2). The persuader is the system designer; designing the system that is to persuade its users. The persuader addresses the overall deliberate behavior or attitude change that resides at the macrosuasion level in which the overall intended change in attitude or behavior is to take place. In reviewing and

categorizing the research work we carefully observed whether the papers offered clear descriptions of C1 or C2.

The event: The event contains the: use, user, and technology sub-contexts. The *use context* (C3) refers to the problem domain dependent features [12] in the form of well-known problems in the domain. The *user context* (C4) includes, for instance, goals, commitments, compromises, and lifestyles. In other words, it refers to the individual application user and also to her individual traits. The *technology context* (C5) refers to the features of the technological platform. Some aspects of persuasion can be bound with the technology itself. For instance, there might be differences between persuasion through mobile phones and through desktop computers in a similar manner as there are differences between persuading in speech and writing. For each research paper, it was noted if authors clearly described the technology context. We ended-up categorizing the papers here with the following technology categories: mobile, desktop, OEM/custom, multiple, ubiquitous, or other. Most web applications were categorized as ‘desktop’, whereas a web application specifically designed for mobile usage was categorized as ‘mobile’. The OEM/custom category was applied to prototype products that were considered unique, for instance an interactive flower pot bowl [20] or an experimental use of a large screen display and a hand controller.

The strategy: The model emphasizes two strategic elements, namely the message and the route. The *message* (C6) refers to the form and/or content selected to deliver the intended transformation; it is the fashion in which the persuader chooses to convey his persuasion for the planned (behavior or attitude) change. The content could be statistical data about the health risks of smoking, but the form might vary as this information could be given to the application user in raw text, it could be built into a dialogue system, or it could be revealed during the course of a game. The persuasive designer in his design attempts to fit with the content and form in such a manner that the user may be persuaded into changing his cognitive stance. The *route* (C7) for persuasion can be direct, indirect, or both [12]. A direct approach would be to provide one or a few solid arguments, whereas an indirect route would rely on a number of facts rather than on one or only a few convincing arguments. A system might also employ both approaches simultaneously, for instance, offering clear rational arguments while employing design patterns which in themselves have been proven persuasive. Another example might be to use an avatar with a specific voice type to present the message, which may make the actual delivery of the content more persuasive. The users might be unaware that of the persuasiveness of the design pattern or the avatar but they would still feel more inclined to be persuaded through the design choices made by the designer.

The PSD Model encompasses many elements that one would expect to find in a communication model for speech, writing, or media in general. If we as researchers wish to embed a message in a system, we need to exhibit awareness of such elements (even if not necessarily using those exact labels). Unless these elements are reasonably clear in disseminations, readers will not be able to determine what actually took place, who was persuaded of what, and by what means.

2.2 The PSD Model: Design Principles

The second leg in the PSD Model focuses on the persuasion itself, describing how an application may be persuasive on an

operational level. The model describes four dimensions. Each of these addresses an aspect of technology-mediated persuasion. These are relatively well-known persuasive elements that may be employed for persuasive system design. The dimensions are primary task support, dialogue support, system credibility support, and social support; they are described in more detail in the [12].

Primary task support: This dimension addresses the target behaviors. For this dimension seven categories are employed: reduction (P1, P to denote principle), tunneling (P2), tailoring (P3), personalization (P4), self-monitoring (P5), simulation (P6), and rehearsal (P7). In the analysis, it was noted which types of primary task support the research work mainly employed, if any, and whether a paper employed more than one design principle.

Dialogue support: This dimension deals with the feedback that the system offers in guiding the user to reach the intended behavior. This dimension also employs seven design principles for providing dialogue support: praise (P8), rewards (P9), reminders (P10), suggestion (P11), similarity (P12), liking (P13), and social role (P14).

System credibility support: In general, credibility (or ethos) is a persuasive element. The PSD model operates with seven design principles for supporting system credibility: trustworthiness (P15), expertise (P16), surface credibility (P17), real world feel (P18), authority (P19), third party endorsements (P20), and verifiability (P21).

Social support: Social support is another greater category that affects the overall persuasiveness of a software system. The model operates with seven design principles for providing social support: social learning (P22), social comparison (P23), normative influence (P24), social facilitation (P25), cooperation (P26), compensation (P27), and recognition (P28).

2.3 Additional variables

In addition to the elements derived directly from the PSD Model, we also registered four additional variables that were deemed relevant in analyzing the papers to reach a finer level of granularity and clarity.

For each of the papers, it was noted whether *ethics* was addressed (A1, A denoting Additional variable). Only papers that discussed ethics at some length were included. Thus, mentioning ethics on the fly in a single sub clause would not count here.

It was also noted whether or not the papers clearly described the *transformation* (A2), if any, that had taken place. For example, did the test subjects rate one condition significantly higher than another, or did the system users’ self-reports clearly demonstrate one design being more persuasive than another?

For each paper, we also registered the meta-data of *contribution type* (A3). This was used to recognize whether the paper was conceptually or empirically oriented. Only papers which provided original data were considered ‘empirical’, whereas papers which borrowed data from previous research as well as theoretical discussions were considered as ‘conceptual’ contributions.

When possible we also noted *the sample size of the subjects* (A4).

3. Results

Different kinds of papers provide very different types of contributions. Experimental papers mainly describe existing

persuasive systems, whereas conceptual papers typically deal with a higher abstraction level and discuss the key concepts behind the systems. Of the 51 papers reviewed the dominant contribution type (A3) was experimental: 62.7% (n=32), whereas 37.3% (n=19) were conceptual. The sample size (A4) of experimental papers varied from 3 to 400, the median being 88. The relatively high median was mostly a result of many surveys in the data.

3.1 Persuasion Context

3.1.1 The intent

The persuader (C1). Quite surprisingly, only 28.1% (n=9) of the reviewed 32 experimental papers clearly stated the persuader so that it was possible to extract ‘who was speaking’. In most cases, it was simply not possible to explicitly recognize the deliberate persuader. Stating that ‘a system was built to...’ does not reveal much about the motives of the speaker and what traits might apply to him in the context of the persuasion itself. Stating that work was financed by a grant from a certain governmental body does not allow us to gain a deeper insight either. A large part of understanding persuasion pertains the persuaders: Who are they? What did they stand to gain from the change? Was it the scientists that build the system themselves or did they act as persuasive system spin doctors preparing a ‘speech’ for someone else?

Change type (C2). Of the 32 experimental papers 81.3% (n=26) stated the change that they were aiming at in a clear manner. Some of these papers did not clearly state their target, but in reviewing the papers it became clear, for instance, that the target was some sort of a behavioral change. Three papers (9.4%) did not state clear change and for three papers (9.4%) it did not apply (experiments that attempted to uncover an aspect of persuasion itself, rather than persuading users of something). Of the same 32 experimental papers 84.4% (n=27) addressed behavioral change and only five (15.6%) aimed at infusing an attitude change.

That behavior was mostly addressed is probably due to the fact that behavior change is in most cases easier to study than attitude change. For instance, in most HCI studies testing two interface conditions can be conducted as clearly quantifiable lab studies within a given timeframe and with a limited budget, whereas the temporal dimension of measuring longer term attitude change makes it more complex and may demand much more complex modes of inquiry. According to Oinas-Kukkonen and Harjumaa [12, 13] persuasion in full happens only when an attitude change takes place.

Optimally, any persuasive systems designer aims to create a system which enables a predictable deliberate transformation of the end-user’s cognitive and/or emotional state. In a sense this offers unique and quantifiable measures of system success. However, the clarity of this depends upon explicitly stating both the aim of the system and the extent to which the system succeeds in doing achieving this aim (and how the success was measured). If, for instance, a researcher states that a system is designed to persuade users who are not exercising currently, into exercising half an hour three times a week, there is quite a clear indicator for measuring the success of the system (assuming that we are informed about the whole context of the persuasion). 81.3% (n=26) of the experimental papers reviewed stated a clear change, thus mentioning what they were trying to accomplish. This may be considered positive, as change and transformation is the core of persuasive systems; however only 71.8% (n=23) of the

experimental papers actually reported whether a change or *transformation* had occurred (A2). Thus some papers stated aiming for a change, but did not state if it actually took place.

3.1.2 The event

Use context (C3). Only 54.9% (n=28) of the 51 papers described the use context. For 19.6% (n=10) it was simply not described whereas for 25.5% (n=13) it was not applicable (papers with conceptual and theoretical content). Even with those papers that described the use context it was often described in a coarse manner. It does not seem to make much sense to simply state that one is designing a system that will resolve or relieve a large and diverse user group’s obesity problems. In comparison, a statement such as: “the goal was to persuade low-income school children between the age of 10 and 15 to eat at least three pieces of fruit each day” seems much sounder way to define a clearer and measurable goal, too. The issue we flag here is that the information going into a design often seems to be at too high a level and detached from an individual user. By hastily jumping from a higher abstraction level to a concrete problem solving level, it may become impossible to see what really takes place through the software system at hand as well as to what extent the change is due to the intervention. Thus, there seems to be a tendency for persuasive system researchers to describe the use context at a very general level, such as the effects of e.g. smoking or malnutrition on society in general.

User context (C4). Of the 51 papers 51.0% (n=26) it was not possible to determine the user context in a clear manner. Only 12 papers (23.5%) addressed the level of *actual users* and *their individual context*. For 13 papers (25.5%) the concept of User Context was not applicable (papers with conceptual and theoretical content). As with the use context most papers discussed also the user context from the higher level of society *à la* “heart disease is an expensive and growing problem in society”. Concrete information was often limited to gender and age and in most cases readers were not supplied with enough details about the subjects’ lifestyles, needs, habits, etc. As an example hereof, in most experimental papers students are seen as one homogenous mass. In most cases readers do not get much information about them other than simply stating that they are students and possibly also their age and gender. There is the danger of oversimplification here. Interestingly, we observed that *systems are often designed to persuade an individual at a system-to-user level but to resolve a user-in-society issue*. Paradoxically, it is not a ‘society’ that is to be persuaded, but an individual user in a society, also it is striking that the designs aimed at a system-to-user transformation are largely informed by system-in-society traits. The designers may however possess in-depth knowledge about users; however it is not clearly reflected in the dissemination. In every case, it becomes difficult if not even impossible to discuss the generalization of results unless a very clear description of the user context has been provided. For the 32 experimental papers, 46.9% (n=15) were conducted with ‘diverse’ subject groups, and 28.1% (n=9) were conducted with ‘students’. 1 paper (3.1%) did not state user context clearly and the rest 21.9% (n=7) were more specific about their user segment e.g. elderly, teens or kids. This might not pose a big problem in a pure human-computer interaction study, where one were testing the users’ cognitive skills, apparatus, or ability to perceive an interface and/or solve concrete tasks with it, but it does pose a problem when designing for persuasion. Persuasion does not

equal raw cognition of buttons, measuring hand-eye coordination or similar. We will have to accept the multidisciplinary nature of the field and observe communication studies, where researchers operate with more finely grained segments. Thus, a clearer research setting could be defined as ‘obese 55-65 year old white females who have higher education’ and such an application audience may be addressed very different than ‘financially challenged uneducated obese teenagers’. In addition, Khaled et al. [23] point out that most research springs from an individualistic western culture as opposed to more collectivist cultures. At this point in the history of persuasive systems we do not know much about how these factors might affect technology-mediated persuasion, but we know that these things play a role in human-to-human communication and in other types of media.

Technology context (C5). In our review of the 32 experimental papers 14 (43.8%) were desktop, 6 OEM/Custom (18.8%), one (3.1%) mobile, and one (3.1%) ubiquitous. Three (9.4%) fall into the category of multiple technologies, and two (6.3%) papers simply do not allow for an extraction of a concrete technological context. Five papers (15.6%) provide qualitative studies in the early phases of investigating a system to be produced at a later time. Naturally, in describing a persuasive system we need a very clear description of the technology context. In spite of the fact that many papers describe the persuasive system under investigation with several screenshots, descriptions of the flow between parts of the system and its content, it is often difficult to grasp what the system really is like. As such, the field faces a challenging task that might be somewhat unique, at least when seen from a hermeneutical perspective, as it simply is hard to communicate a user experience very clearly. One may be able to ‘feel’ an interface and the actual system when interacting with it, whereas one does not ‘feel’ the physical word set in ink on paper when one reads it. Rather we may feel the *meaning* that we interpret those words to convey – but where does this meaning reside in a persuasive system? If one subscribes to Redström’s (2006) notion that persuasive system artifacts de facto have embedded arguments within them [14] and extend this view to encapsulate the rhetorical notion that an argument might be an appeal to either emotions (ethos and pathos) rather than logic (logos), we can begin to sense how complex the description of technology context can be. What, after all, caused the actual persuasion? Was it the content of a game that was persuasive, or was it the presentation layer? Was there really just one type of content? Was it the smooth interaction (the feeling of clicking and navigating and receiving feedback), or was it a mix of all of these concepts? Notably, when discussing persuasive systems we should remember that while the technology in itself is not neutral [12, 14] it is not persuasive either. An example might clarify this important point: When conveying the message that attending sports three times a week is essential for schoolchildren’s health (and persuading them into changing their behavior in that direction), we could choose to implement two different systems. One could be a mobile application promoting physical activities for the children and could be an information kiosk to be deployed at schools near the canteen. Notably, only the use context and the technology context differ between these two systems. If at a later time one would then survey the utilization of these systems, one could wind-up determining that the success seems to be higher when deploying a mobile application; that does not, however, mean that the mobile

applications *per se* are more persuasive than information kiosks. They may, for an instance, be more persuasive for domain-specific issues and a user group at hand; but not more persuasive as such.

3.1.3 Strategy

The message (C6), comprising both the form and the content to cause an intended transformation, was described in 62.5% (n=20) of the 32 experimental papers. It was not described in 28.1% (n=9), and for the remaining 9.4% (n=3) it did not make sense to distinguish between these two, for instance a qualitative analysis on a topic that would affect communication and thus rather describe an aspect of content than delivering content themselves. Many of these papers were measuring the difference between two or more conditions rather than conveying a message as such.

The route (C7). In general we were not able to extract the route taken for persuasion. In the 28.1% (n=9), of the papers that did not offer a clear message (C6) it was naturally not possible. In papers that did have a clear message, it was not described at a granularity; where we could determine the route clearly. Typically the message is not described at such a level of detail that we can determine whether they had employed a direct approach thus accomplishing the change through a convincingly strong argument or whether they rather employed many arguments. We found that too much was left to interpretation when categorizing papers route for us to draw any real conclusions from. This is in itself a finding as it again highlights that the dissemination of persuasive system research offers added complexity. We need to know not only the message but also what *kind* of route the message takes: What types of arguments were presented? Was one or multiple arguments presented? Possibly also the types of rhetorical appeals employed (logos, ethos, pathos or a mix?). When this information is omitted, it becomes hard for readers to understand what took place and what caused the actual persuasion. Was the system e.g. manipulative or was it pure voluntary change?

3.2 Design Principles

Many papers used multiple ways to support the persuasion and oftentimes they used multiple techniques to do it. The most used design principles tackled in the data can be seen in Table 1.

Tailoring, tunneling, and reduction seem to be the most used ways for persuasion to support accomplishing one’s primary task. On the other hand, suggestion as way to support the user-system dialogue and social comparison and normative influence as a means to provide social support seem to have received much attention in research so far. From the credibility research, surface credibility has received most of the attention so far.

Persuasion technique	Support dimension	Frequency
Tailoring	Primary task	11
Social comparison	Social	11
Tunneling	Primary task	10
Reduction	Primary task	10
Suggestion	Dialogue	9
Surface credibility	System credibility	8
Normative influence	Social	7
Self-monitoring	Primary task	6
Social learning	Social	6
Praise	Dialogue	5
Liking	Dialogue	5
Simulation	Primary task	4
Reminders	Dialogue	4
Authority	System credibility	4
Recognition	Social	4
Rewards	Dialogue	3
Similarity	Dialogue	3
Trustworthiness	System credibility	3
Cooperation	Social	3
Personalization	Primary task	2
Rehearsal	Primary task	2
Social role	Dialogue	2
Expertise	System credibility	2
Real-world feel	System credibility	2
3 rd party endorsements	System credibility	2
Social facilitation	Social	1
Verifiability	System credibility	0
Competition	Social	0

Table 1. Most used design principles in the Persuasive conferences 2006-2008.

3.3 Ethics

Surprisingly, ethics (A1) was hardly addressed in the data as only three of the 51 reviewed papers discussed the topic at length. None (!) of the experimental papers explicitly addressed ethical considerations. This is a severe shortcoming in the current state of art in the field. Ethics being largely unaddressed might in itself be considered unethical, if the field were to boast its concern about the ethical aspects of computer-mediated persuasion.¹ Moreover, there are some large unaddressed questions to be discussed. To begin with, does the field truly subscribe to the notion that persuasive systems should cause a *voluntary* cognitive change? If so this is unfortunately not clearly visible in the reported research. We must clearly state that designing persuasive systems undertake a new meaning in particular with regard to the ethical impact. Persuasive system designers can never assume that they are merely building software tools, for in doing so they are also deliberately conveying embedded messages with the aim of

infusing a planned cognitive change in the end-users. This is only to a lesser degree the case when we design reformative systems, if for instance we build a phone system supporting speech transmission; we are not ethically responsible for one user calling another user and issuing a threat. If we however build a phone system that somehow persuades such behavior that would naturally pose severe ethical implications. Persuasive designers undertake the task of embedding deliberate messages in persuasive systems by designing the arguments that are contained within the systems [14] resulting in systems that persuade on our behalf – systems that act as our vehicle of persuasion. Therefore, to a high degree, we must assume responsibility for the ethical aspects of such designs as we embark on encouraging a certain behavior or attitude [8, 15, 16]

3.3.1 Bias and validity

When conducting an analysis such as described here, potential bias lies in the interpretation of the published research papers. In reviewing and categorizing, we carefully observed if the authors clearly stated the described variables. Naturally, the papers did not always employ the very same terminology as found in the PSD Model, so in many cases the analysis was based on an interpretive categorization. According to Andrew, Borriello and Fogarthy, persuasive strategies in themselves may overlap [21], which naturally also hints that interpretation in some cases might be necessary. Interpretation is always partly subjective [22]. We did run into this issue, however in most instances it was relatively easy to categorize as we employed the clearly defined elements of the framework. Theoretical papers did not necessarily describe one specific system, but rather described concepts in a more general fashion. Both papers that describe tunneling as a strategy from a theoretical perspective and papers that describe a system actually employing tunneling were noted in the tunneling category for primary task support. This ensured that the model yielded meaningful data for papers that were not founded in an experiment but still presented in-depth knowledge about a given topic.

4. Conclusions

This paper has provided an overview of the current state of the field of persuasive system design. The field has had a good start and it may evolve into something extraordinary but there is room for improvement in both the research topics and methods utilized. One of the main findings in this paper is that all too often there are blanks in the dissemination work itself. We believe that this is due to the inherently multidisciplinary nature of persuasive systems as well as the limited age and relative immaturity of the field. As a preliminary contribution we have attempted to devise a heuristic (included in Appendix 1) that addresses the most central issues to keep in mind in dissemination. The heuristic may serve as a starting point when embarking on dissemination work.

More is required from the systems when they are to persuade their users, but equally important more is also required from the system designers. The skill sets that a persuasive designer optimally should maintain requires expertise from multiple diverse areas; which leads to an increase in complexity. Persuasive designers will have to know about many different aspects of technology, and at the same time to understand the deliberate content as seen from the perspective of persuasive communication, i.e. rhetoric. This should be reflected in the educational programs on persuasive systems and design. One cannot be an expert in all of

¹ It should be noted, however, that at Persuasive 2008 there was a panel session dedicated on ethics of persuasive technology, and at Persuasive 2007 it was raised as an issue in one of the panels.

these diverse areas, but it seems clear that there should be a common body of references from each of the fields stipulating their contribution to persuasive systems.

5. Directions for Future Research

Based on the findings we will also draw more general conclusions and provide suggestions for future research directions. The suggested themes are in no ranked order and are by no means exclusive or exhaustive. They merely stipulate some of the findings and hint at possible solutions.

Persuasive design methods: There will be a continuous need for research into how to *prescribe* predictable persuasive designs and also methods for a clearer measurement of successful designs. As of now, there are not many conceptual models or persuasive system design methodologies. Empirically proven models would be of great value for the field.

Persuasive design patterns: Some modes of interaction are in themselves more persuasive than others. By properly designing options and visual feedback we may devise optimal sequences of interaction that cater more to a certain type of behavior than another, for instance, designing a set of web application screens with the aim of having people signing up to a news mail, adding more things to their shopping cart, or inviting their friends to engage in an online community. Where we in HCI might focus on how to make such task solving as easy as possible and ensuring that we have the least amount of cognitive strain, persuasive interaction design patterns caters to the users on a different level formalizing how we might get more success in persuading the user to complete such tasks.

Software audiences: The notion that people are both users and audience needs to be explored further and at depth. There might well be a need to target the design of systems depending on the audience that we are actually trying to persuade. If, for instance, we are to persuade young males in urban areas not to drive their cars too fast, chances are that we would need to tailor or message very clearly towards them in the persuasive system. What is special about persuading young men? It is possible to point out a set of specific target audiences that might be addressed on their own, e.g. kids, tweens, teens, young adults or the elderly. We should also note such gender and cultural differences that might play a role in persuasion.

Scientific and theoretical concerns: The multiple scientific areas have different epistemological traditions. This in combination with the broad nature of persuasion results in a multi-disciplinary area of investigation open to a wide array of themes as well as approaches. Regardless, we must come to terms with how we can combine engineering with a field that has a two thousand years old tradition of dealing with ontology and debating the true nature of meaning as such.

Ethical concerns: The field has faced and will continue to face ethical challenges as, no doubt, many research findings from the field of persuasive systems can be abused. Ethical considerations should be more clearly addressed in the research. Unfortunately, our review revealed that so far these have been poorly addressed in the published research papers. Special attention should be given to situations where computer-mediated persuasion takes place without the user being aware of it [15]. Also, the 'grey areas' should be considered, e.g. would it be unethical to design manipulative systems that would lead elderly users into taking

their heart medication in the most optimal way. In spite of potentially noble outcomes, there may be many ethical aspects to be debated. The field of persuasive system design should take upon itself also to monitor and share examples of unethical behaviors, for instance documenting unethical marketing and/or propaganda.

Application domains: *e-Health* as an application domain is both so large and important that it deserves special attention. Many of the health challenges faced with in the western society can only be solved by infusing humans with the motivation to make long-lasting lifestyle changes. For instance, obesity, alcoholism, internet addiction, compliance and corrective behavior technologies, social support, and digital interventions provide a multitude of interesting research questions. Designing persuasive systems that could resolve even some small parts of these problems and aid in true long-term sustainable change would provide to be very valuable. *Knowledge work and collaboration* is another area of its own that earns merit. Admittedly, this is a very challenging area. Whether to address it at a microsuasion level, i.e. in user recommendations in the form of rating, or whether to aim at designing systems that would persuade users to engage in more knowledge sharing, there is a need to investigate how humans can be persuaded into undertaking significant feats. Currently, Wikipedia is a prime example of true collective intelligence. We could also mention open source software projects, such as Linux, in which users are eagerly contributing valuable knowledge. As knowledge work and the augmentation of the human intellect by means of collective intelligence become more prevalent, there is a need to be able to design systems that may persuade users to engage in such altruistic behaviors.

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7. Appendix 1: A heuristic for persuasive system research dissemination.

Below we offer a simple heuristic that emerged from our work with the literature review. It reflects variables that make dissemination in regards to persuasive systems clearer, as their explicit mentioning aid the reader in understanding how the

persuasive system worked. In system descriptions, most of the variables can be made explicit relatively easily:

1. Persuader
 - a. Who is speaking via the design? Who embedded an argument in the artifact?
 - b. Why are they speaking? What is their intention or purpose?
 - c. What do they stand to benefit from successful persuasion?
 - d. What is their background and culture, etc.?
2. Change type
 - a. What is the goal? Behavior and/or attitude change?
 - b. What transformation should the software system produce?
 - c. What other approaches have failed?
 - d. Did the software system finally produce the desired transformation?
3. Use context
 - a. Who are the users as a group, e.g. socially challenged girls 10-12 or 55 years and older?
 - b. What problem-domain dependent features in the form of well-known problems are to be addressed by the design? Physical, cultural/normative, or competitive?
 - c. Who (or what) else is competing for attention in this space?
4. User context
 - a. What is specific for the users with regard to what they are to be persuaded of?
 - b. Why is there a need for persuasion? What constrains their decision?
5. Technology context
 - a. Single hardware platform or multiple?
 - b. Networked or stand-alone software?
 - c. Single or multiuser software, or shared collaborative?
6. Message
 - a. What is the form? Why was it chosen? What is the genre, e.g. game, social community, tracking tool, ambient system, etc.?
 - b. What type of content is the system providing?
 - c. What kind of appeal is mostly employed? Logos, Ethos, Pathos or Mix?
 - d. What kinds of arguments are being provided?
7. Route
 - a. What is the approach? Direct (one dominant argument) or indirect (several arguments)?
8. Experimental papers
 - a. Did the system *measure* a degree of persuasiveness between conditions or did the system *actively persuade* of something?
 - b. Number of subjects
 - c. Method employed
 - d. Analysis of results
 - e. Level of transformation
9. Ethical considerations and implications